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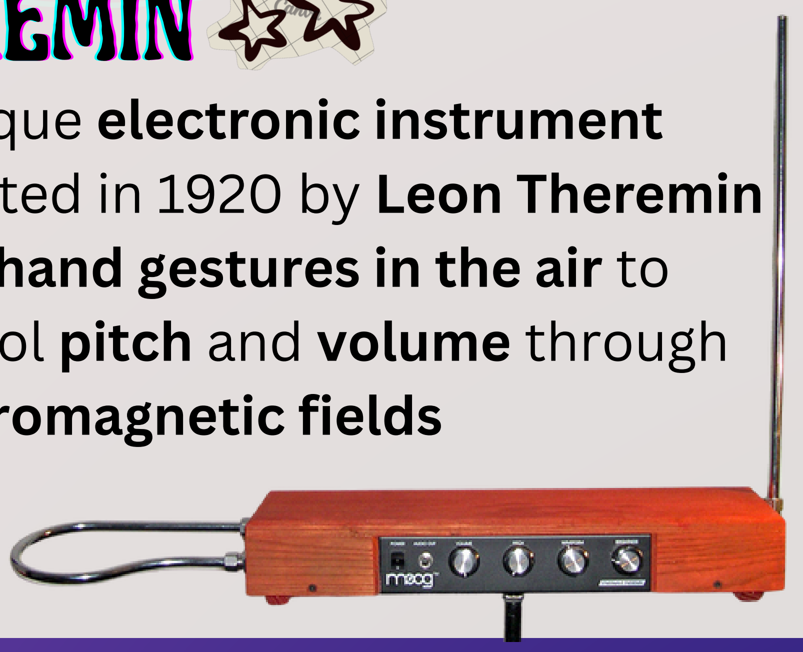
2026 PHYSICS WRAPPED

THIS YEAR YOU LISTENED TO 50,000HZ OF...

THEREMIN

- A unique **electronic instrument** invented in 1920 by **Leon Theremin**
- Uses **hand gestures in the air** to control **pitch** and **volume** through **electromagnetic fields**



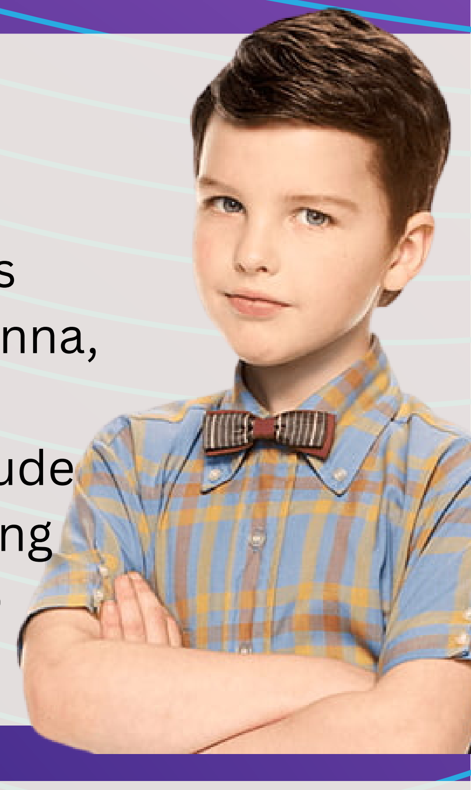
ELECTROMAGNETIC WAVES

Pitch Control

- **Capacitance forms** between the hand & antenna, causing **electric fields** to be formed
- Position of hand **changes the effective distance** and thus **capacitance**, eventually changing the **frequency**, producing different **pitches**

Volume Control

- When hand moves closer to the antenna, **RF energy is absorbed**, amplitude decreases, resulting in a **lower volume** produced



MUSICIAN TECHNIQUES

Stance

- Stand roughly **0.3m** away
- **Dominant foot** in front and **towards pitch antenna**

Tuning

- **Lowest note** to be played at the **musician's right shoulder**

Playing

- For pitch, **fully open fingers** usually represent the **8th** position (e.g. High C) and **fully curled up fingers** represent the **1st** position (e.g. Middle C)
- **Fine control** of notes give distinct, rich sounds
- **Clear ~3m** of space around the theremin to prevent interference with the EM waves

BONUS:

Scan to watch Leon Theremin play his own instrument.

Scan to watch Sheldon play the theremin on the Big Bang Theory.

THE SCIENCE BEHIND IT

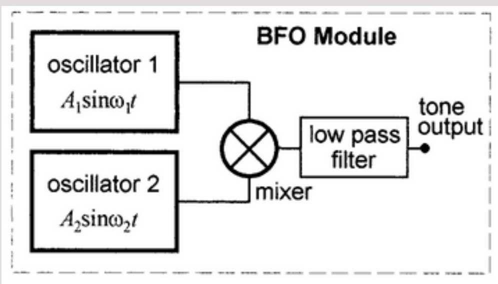
Beating

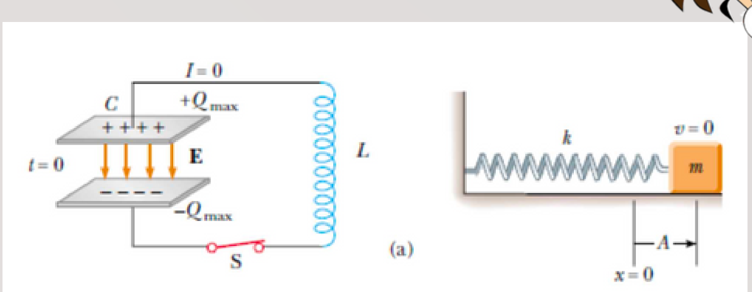
- The **audio signal** includes **two frequency sources** with one **fixed** and one **variable**
- The two sources are **fed through a mixer** which **multiplies the signal**, creating a **beating wave**

$$q(t) = A_1A_2\cos(\omega_1 t)\cos(\omega_2 t) = \frac{A_1A_2}{2}(\cos((\omega_1+\omega_2)t) + \cos((\omega_1-\omega_2)t))$$

- Beating wave goes through a **low pass filter** which **filters out the high frequency term**, leaving us with:

$$q_{filter} = \frac{A_1A_2}{2}\cos((\omega_1-\omega_2)t)$$





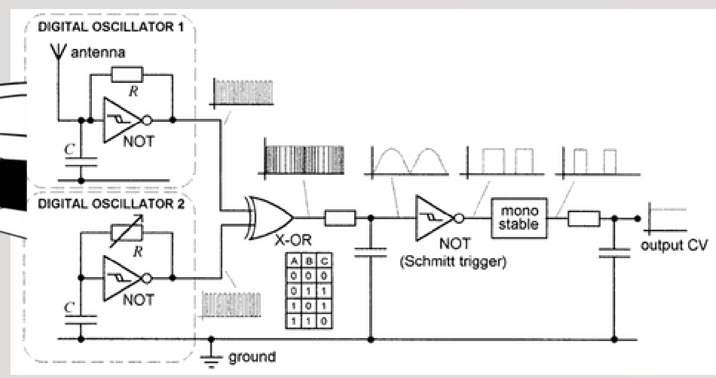
Frequency Control

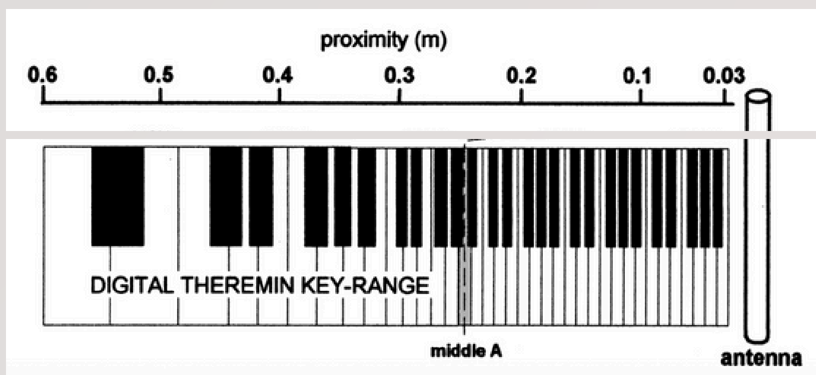
- **Oscillators** function like a **simple LC circuit**
- Can be thought as a **mass-spring system**, with **inductor L as inertial (mass) term** and **capacitance** $1/C_{tot} = 1/C_{in} + 1/C_A$ related to the **stiffness term**
- The differential equation satisfies $L\frac{d^2}{dt^2}I(t) + \frac{1}{C}I(t) = 0$
- Oscillator frequency is dependent on the antenna capacitance, which is grounded by the human

$$\omega_0^2 = 1/LC = \frac{1}{L}\left(\frac{1}{C_{in}} + \frac{1}{C_A}\right)$$

Volume Control

- **Uses an oscillator similar to the frequency control**
- The output is sent to a **xor gate**, which makes the rising edge appear at the **frequency of ~the sum of oscillator freq.**
- After smoothing, it outputs DC that is **related to the capacitance of the volume antenna**





OTHER APPLICATIONS OF ELECTROMAGNETIC WAVES

X-ray

Visible Light

Microwave

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